

mbusgw-t2r

User Guide

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NaPi World & StarLet Squad collaboration

net2serial User Guide

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Preface

1. About This Manual

This manual describes the configuration, the operation and the troubleshooting of net2serial --- a gateway which passes a raw octets stream between a TCP connection and a local serial port, in both directions and without interpreting it.

2. Intended Audience

This manual is written for a reader with no prior serial line or Linux administration experience: every step is spelled out and nothing is assumed. An experienced engineer may go straight to Chapter 3 (the settings file) and Chapter 6 (the troubleshooting).

3. Document Structure

Chapter 1 explains what the gateway does and why a port is given to a single client. Chapter 2 is the pre-flight checklist. Chapter 3 describes the settings file key by key. Chapter 4 covers the start, the command line options and the first test. Chapter 5 teaches how to read the log. Chapter 6 is the troubleshooting reference: the symptom, the cause, the action, and the reference of all the message codes.

4. Related Documents

README.md --- the installation sequence (the StarLet package, the build, make install). The manual pages of `termios(3)` and `stty(1)` --- the serial line parameters of the operating system.

5. Conventions

Convention	Meaning
Monospace type	Code examples, file names, commands and interactive screen displays.
%N2S-E-CODE	A message code as it appears in the gateway log; see Section 6.2.
<...>	A placeholder to be replaced by an actual value, e.g. a device name.
\$	The shell prompt of an unprivileged user; do not type it.
Ctrl/C	Hold down the key labeled Ctrl while you press the key C.

Chapter 1. What the Gateway Does

You have a device with a serial console or a serial interface --- a switch, a router, an old VAX, a PLC, a UPS. It is plugged into the serial port of the computer where this gateway runs. And you want to reach that device over the network, from your desk. The gateway sits between them and passes the octets through, as Figure 1 shows.

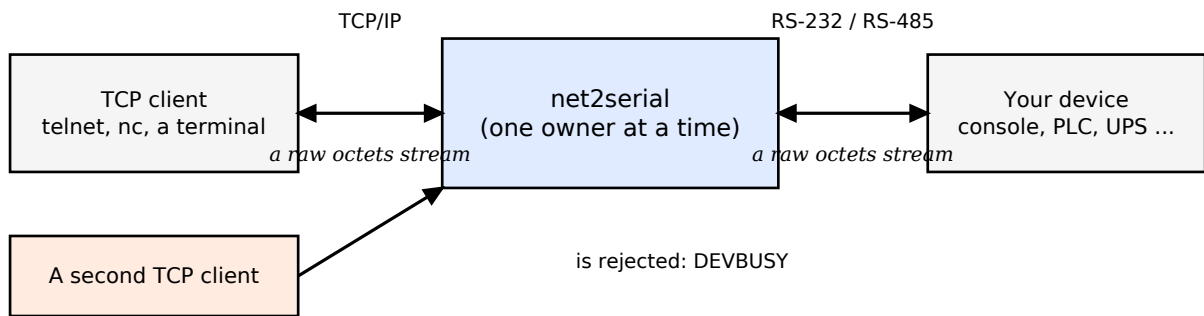


Figure 1. The data path of the gateway

Nothing is interpreted: what you type goes to the device as is, what the device prints comes back as is. This is why it works with anything --- a login prompt, a bootloader, a binary protocol.

1.1. One Client at a Time per Port

An octets stream has no framing. If two people typed into one console at once, their characters would interleave and neither would get a readable answer. So a serial port is given to a single session: a second client is refused with a clear message and the first one keeps working undisturbed. As soon as the first client disconnects, the port is free for the next one --- see Figure 2.

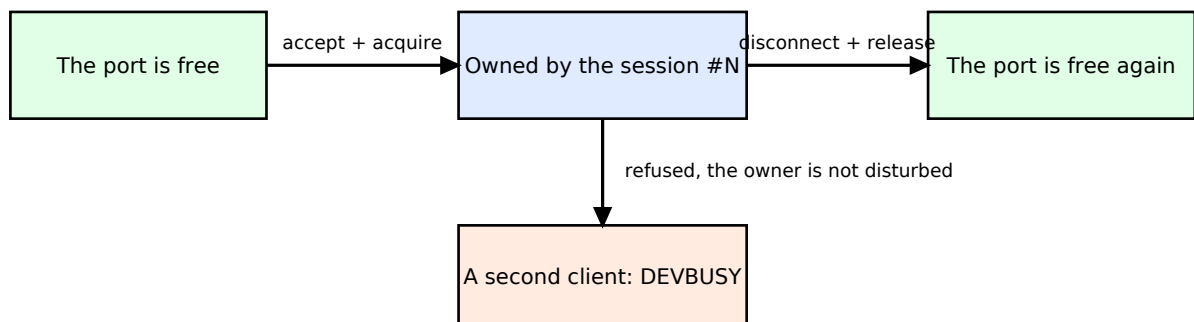


Figure 2. The lifecycle of the port ownership

Chapter 2. Before You Start

Walk this checklist once --- it saves an hour of guessing later.

1. The gateway is installed (see README.md). After the installation you have the program `/usr/local/sbin/net2serial` and the settings file `/usr/local/etc/net2serial/net2serial_settings.conf`.

2. You know which serial port your device is connected to. On Linux it is a file like `/dev/ttyS0` or `/dev/ttyUSB0`. If unsure, plug the USB adapter out and in, then run:

```
$ dmesg | tail
```

3. You know the line parameters of the device: the speed, the data bits, the parity, the stop bits and the flow control. They are in the device manual; the most common console set is 9600, 8, N, 1 with no flow control.

4. Your user may open the port. Check the group of the port file:

```
$ ls -l /dev/ttyUSB0
crw-rw---- 1 root dialout 188, 0 ... /dev/ttyUSB0
```

If the group is `dialout`, add yourself to it and re-login (or simply run the gateway with `sudo`):

```
$ sudo usermod -a -G dialout $USER
```

Chapter 3. The Settings File

The settings file has two sections: `serials` (the serial ports) and `listeners` (the TCP ports). A minimal working example:

```
serials = (
  {device = "/dev/ttyUSB0";
  chars = "9600, 8, N, 1";
  flow = "NONE";
  }
);

listeners = (
  {bind = "TCP:0.0.0.0:5000";
  target = "/dev/ttyUSB0";
  }
);
```

Read it as: "open /dev/ttyUSB0 at 9600-8-N-1 with no flow control; listen for TCP clients on every interface, port 5000, and connect them to that port".

3.1. The serials Section

Key	Required	Meaning and the allowed values
device	yes	The serial port file, e.g. /dev/ttyUSB0.
chars	yes	The line parameters: speed, data bits, parity, stop bits. Speed 50..4000000; data bits 5..8; parity N (none), E (even), O (odd); stop bits 1..2.
flow	no	Flow control: NONE, XON/XOFF, RTS/CTS. Default NONE.
iotmo	no	The I/O timeout, milliseconds, 1..600000. Default 3000.
rs485	no	1 --- ask the kernel to drive the RS-485 direction control. Default 0.
desc	no	A free text description, for your own convenience.

Which flow control to pick? If the device manual says nothing --- NONE. A Cisco-style console cable usually wants RTS/CTS. Old terminals and printers often want XON/XOFF. If you paste a long text and the tail comes out garbled, the flow control is the first thing to check.

If a record is wrong, the gateway skips it and says why --- with the allowed range right in the message:

```
%NET2SER-E: [serial #00:</dev/ttyUSB0>] --- speed 31 baud is out of range [50..4000000]
```

3.2. The listeners Section

Key	Required	Meaning and the allowed values
bind	yes	Where to listen: TCP:<IP address>:<port>. The port is 1..65535; 0.0.0.0 means every interface. UDP is not supported.
target	yes	Which serial port to connect to. Must match a device from serials exactly, character by character.
connlm	no	The backlog of the listening socket, 1..128. This is NOT a count of simultaneous clients --- see Section 1.1.
iotmo	no	The network I/O timeout, milliseconds, 1..600000.

Ports below 1024 need the root privileges on Linux. If you do not want to run as root, take e.g. 5000 and point your client there.

Chapter 4. Running the Gateway

```
$ /usr/local/sbin/net2serial /settings=/usr/local/etc/net2serial/net2serial_settings.conf
```

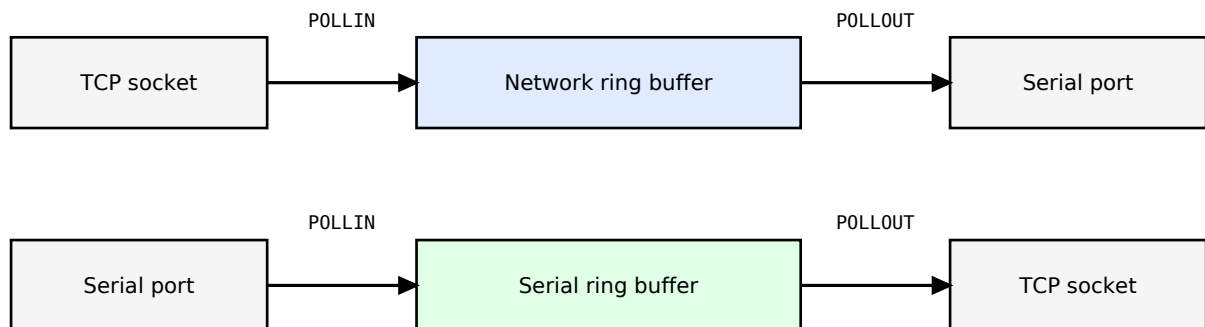
Option	Meaning
/settings=<file>	The settings file (Chapter 3).
/trace	The verbose tracing: every chunk of octets is reported. Priceless at the first run.
/logfile=<file>	Write the log to a file instead of the screen.
/logsize=<octets>	Rotate the log file above this size.

A healthy start prints (shortened):

```
%N2S-I-REVISNF, Rev: NET2SER X.00-05/aarch64(built at ...) (REV: 00.05.00)
%NET2SER-I: Added device #00 [</dev/ttyUSB0>, Chars: <9600, 8, N, 1>, Flow: <NONE>, ...]
%NET2SER-I: Added listener #00 [Target: </dev/ttyUSB0>, Net: <TCP:0.0.0.0:5000>, ...]
%N2S-S-DEVREADY, Device </dev/ttyUSB0> [9600 baud, 8N1, flow: NONE] --- is ready
%N2S-S-LSNRRDY, [#3] Listener 0.0.0.0:5000 [Target: </dev/ttyUSB0>] --- is ready
```

Two lines matter most: DEVREADY (the port is open) and LSNRRDY (the TCP port is listening). If you see both, the gateway is up.

Inside, the two directions are decoupled by a pair of ring buffers, and the poll() events are derived from their state at every round --- this is what gives the natural back pressure to a fast side of the pair (Figure 3).



A free space in a ring -> we ask for POLLIN on its source

A data in a ring -> we ask for POLLOUT on its destination

Figure 3. The ring buffers and the derived poll() events

To stop the gateway press Ctrl/C once and wait a second. To toggle the tracing of a running gateway without a restart:

```
$ kill -USR1 <pid>
```

4.1. The First Test

```
$ telnet 127.0.0.1 5000
```

Press Enter a couple of times --- a console device usually answers with its prompt. nc works too and is friendlier for the binary data:

```
$ nc 127.0.0.1 5000
```

Chapter 5. Reading the Log

Every important event is one line with a code:

`%N2S-E-DEVOPNERR, Cannot open the device </dev/ttyUSB0>, errno: 2`

the facility the severity: S I W E F the message code (Section 6.2)

Figure 4. The anatomy of a log message

The severity letter is: S --- success, I --- information, W --- warning, E --- error, F --- fatal.
Grep the log by the code --- this is exactly why the codes exist:

```
$ grep DEVBUSY /var/log/net2serial.log
```

Chapter 6. Troubleshooting

The golden rule: run with /trace and read the log. The gateway always says what it dislikes.

6.1. The Symptom, the Cause, the Action

You see	It means / what to do
%N2S-E-DEVOPNERR, ... errno: 2	The port file does not exist: the adapter is unplugged or the name is wrong. Run dmesg tail after plugging it in; fix device in the settings.
%N2S-E-DEVOPNERR, ... errno: 13	Permission denied. Run with sudo, or add yourself to the dialout group.
%N2S-E-DEVOPNERR, ... errno: 16	The port is busy: another program holds it. Find it: sudo fuser /dev/ttyUSB0.
%N2S-W-DEVBUSY, ...	Somebody is already working with that port. This is by design (Section 1.1). Wait, or ask the colleague to disconnect.
%N2S-E-LSNRERR, ... errno: 98	The TCP port is taken: a second gateway instance or another program. Find it: sudo ss -tlnp grep 5000.
%N2S-E-LSNRERR, ... errno: 13	The ports below 1024 need root. Run with sudo, or take a port above 1024.
%N2S-E-LINKDOWN, ...	The serial port has died under a live session --- almost always an unplugged USB adapter. Plug it back and reconnect.
%N2S-W-SESSTMO, ...	The session was idle for 20 minutes and was closed. Only truly idle sessions are closed --- any traffic resets the timer.
You connect, but nothing comes back	1) The wrong speed or parity --- re-check chars; 2) the wiring (a null-modem cable is often needed); 3) the device is off or has nothing to say --- press Enter.
The output is garbage characters	Almost always the wrong speed. Try 9600, 19200, 38400, 115200 in turn.
A long paste comes out truncated	The device cannot keep up: set flow = "RTS/CTS" or "XON/XOFF" per the device manual.
... out of range [a..b]	A settings value is out of its range; the allowed range is right in the message.
No serials has been defined!	Not a single serials record survived the validation. Read the error lines above --- each rejected record says why.

6.2. The Message Codes Reference

Code	Sev	When it appears
REVISNF	I	At the start: the program version. Quote it when asking for help.
DEVREADY	S	The serial port is open; the actual line parameters are shown.
LSNRDLY	S	The TCP port is listening; the address, the port and the target device are shown.
DEVOPNERR	E	The serial port cannot be opened; errno says why (2 = no such file, 13 = the permissions, 16 = busy).

Code	Sev	When it appears
LSNRERR	E	bind()/listen() failed for a TCP port; errno says why (98 = taken, 13 = the privileges).
NETCONN	S	A client has connected; its address:port and the listener are shown.
NETDISCN	S	A client has disconnected; its address:port is shown.
DEVBUSY	W	A client was rejected: the port is already owned by another session.
SESSTMO	W	A session was closed after being idle for too long.
NETIOERR	E	A network I/O error; the failed call and the errno are shown.
TTYIOERR	E	A serial I/O error; the failed call and the errno are shown.
LINKDOWN	E	The serial line failed under a live session (an unplugged adapter).
EXITST	I	The gateway exits; the exit flag and the final status are shown.

6.3. If Nothing Helps

Collect and attach to your question: 1) the full start-up log with `/trace` (from REVISNF to the first error); 2) your settings file; 3) the output of `ls -l <device>` and `dmesg | tail -20`; 4) the exact model of the device and its documented line parameters. With these four things the problem is almost always visible at a glance.